

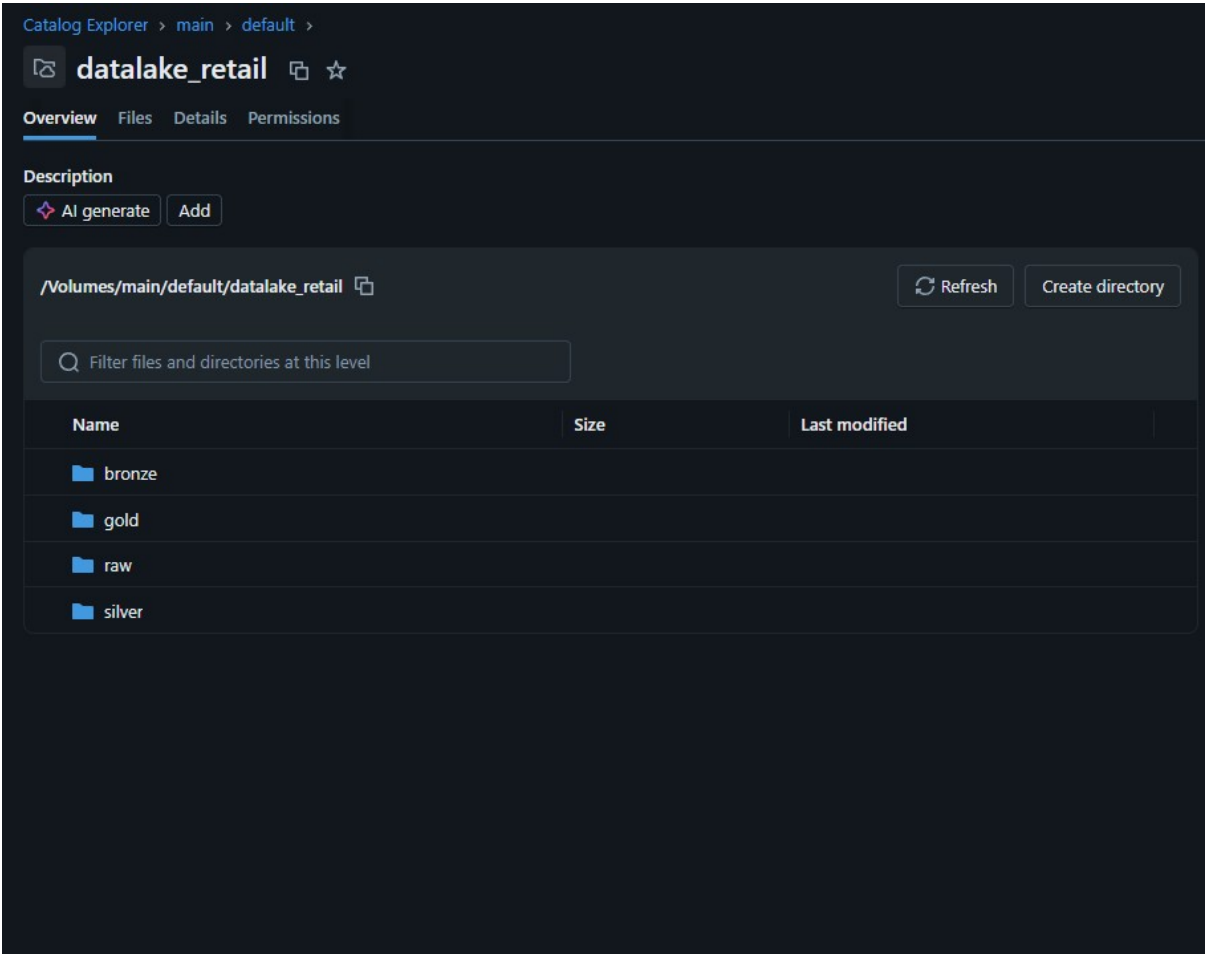
Final Document

Storage Organization (Bronze, Silver, Gold)

The project follows a structured data lake layout using **Databricks Volumes** for physical storage and **Unity Catalog tables** for governance and lineage.

Storage Layout

- **Bronze:** Raw ingested data
- **Silver:** Cleaned and validated data
- **Gold:** Aggregated analytics data



Catalog Explorer > main > default >

 **datalake_retail**  

Overview

Files

Details

Permissions

Description

 AI generate

Add

/Volumes/main/default/datalake_retail / **bronze** 

 Refresh

Create directory

 Filter files and directories at this level

Name	Size	Last modified
 products		
 sales_transactions		
 stores		

Catalog Explorer > main > default >

 **datalake_retail**  

Overview

Files

Details

Permissions

Description

 AI generate

Add

/Volumes/main/default/datalake_retail / **silver** 

 Refresh

Create directory

 Filter files and directories at this level

Name	Size	Last modified
 products		
 sales_quarantine		
 sales_transactions		
 stores		

Catalog Explorer > main > default >

 **datalake_retail**  

Overview

Files

Details

Permissions

Description

 AI generate

Add

/Volumes/main/default/datalake_retail / **gold** 

 Refresh

Create directory

 Filter files and directories at this level

Name	Size	Last modified
 daily_sales_summary		
 monthly_revenue_region		
 product_performance		

Gold Tables in Unity Catalog

Gold datasets are registered as **managed Unity Catalog tables** in main.default, enabling governed access and BI integration.

Gold Tables Created

- daily_sales_summary
- monthly_revenue_region
- product_performance

Data

Search

▼  `main`.`default`.`daily\_sales\_summary`

☐ Σ daily_revenue

☐ Σ total_quantity_sold

☐ Σ transaction_count

☐ transaction_date

▼  `main`.`default`.`monthly\_revenue\_region`

☐ Σ month

☐ Σ monthly_revenue

☐ region

☐ Σ year

▼  `main`.`default`.`product\_performance`

☐ category

☐ product_id

☐ product_name

☐ Σ total_revenue

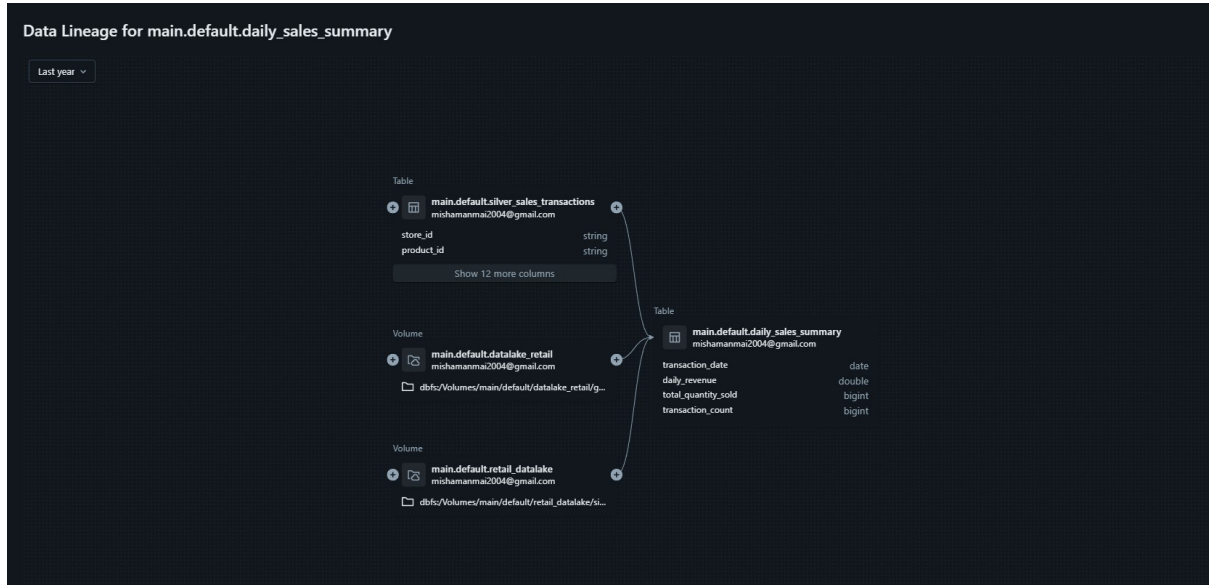
☐ Σ total_units_sold

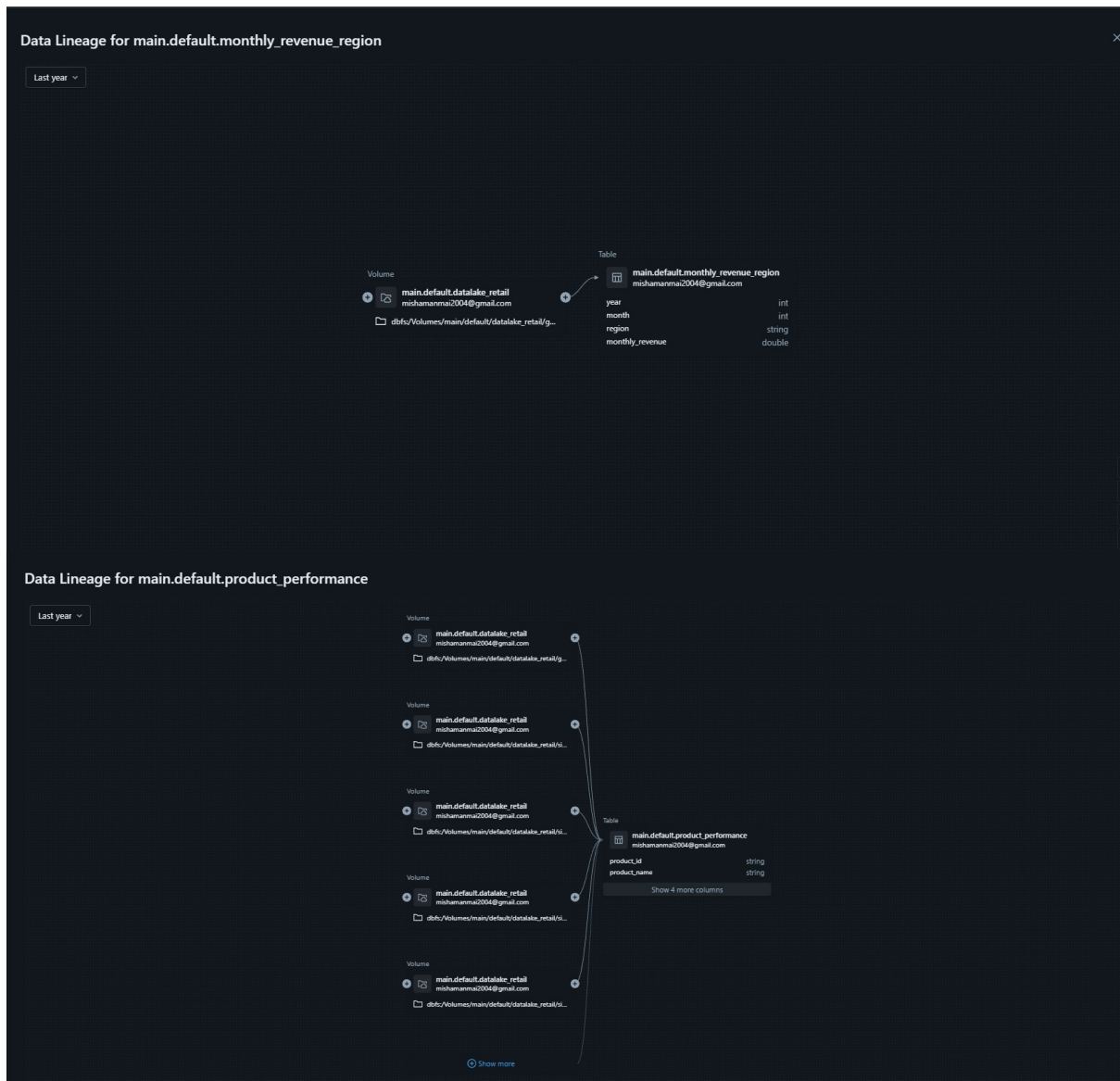
☐ Σ transaction_count



Gold tables are created using **SOL-based table-to-table transformations** from Silver tables.

Lineage Validation



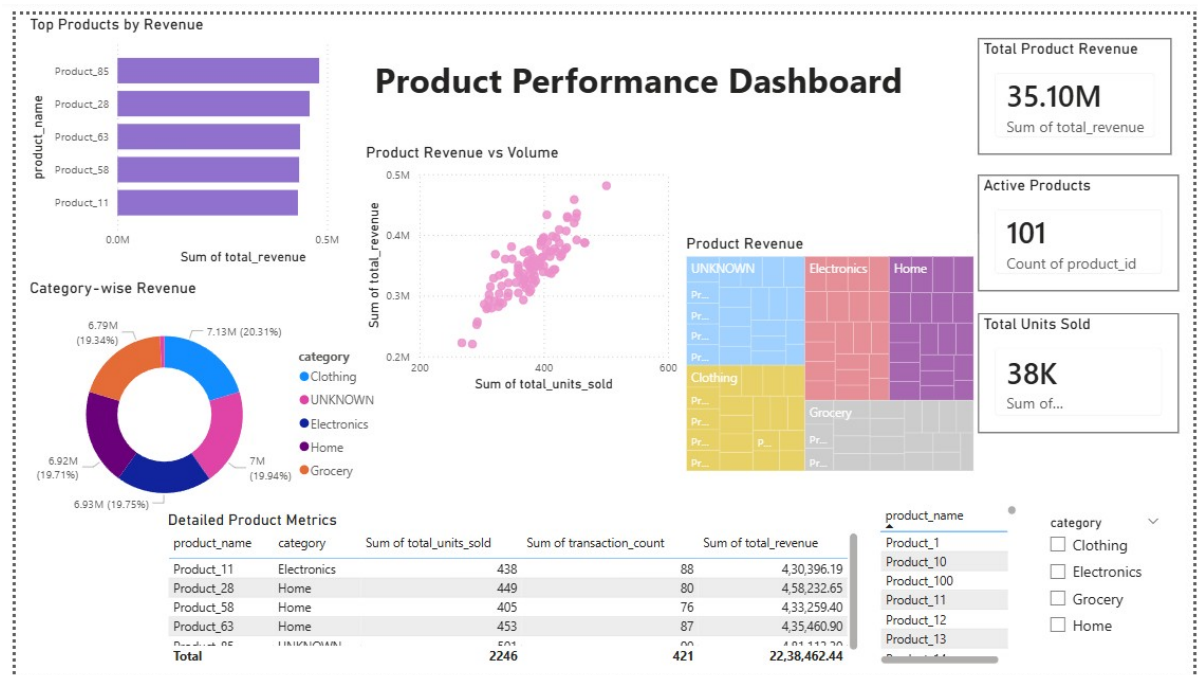
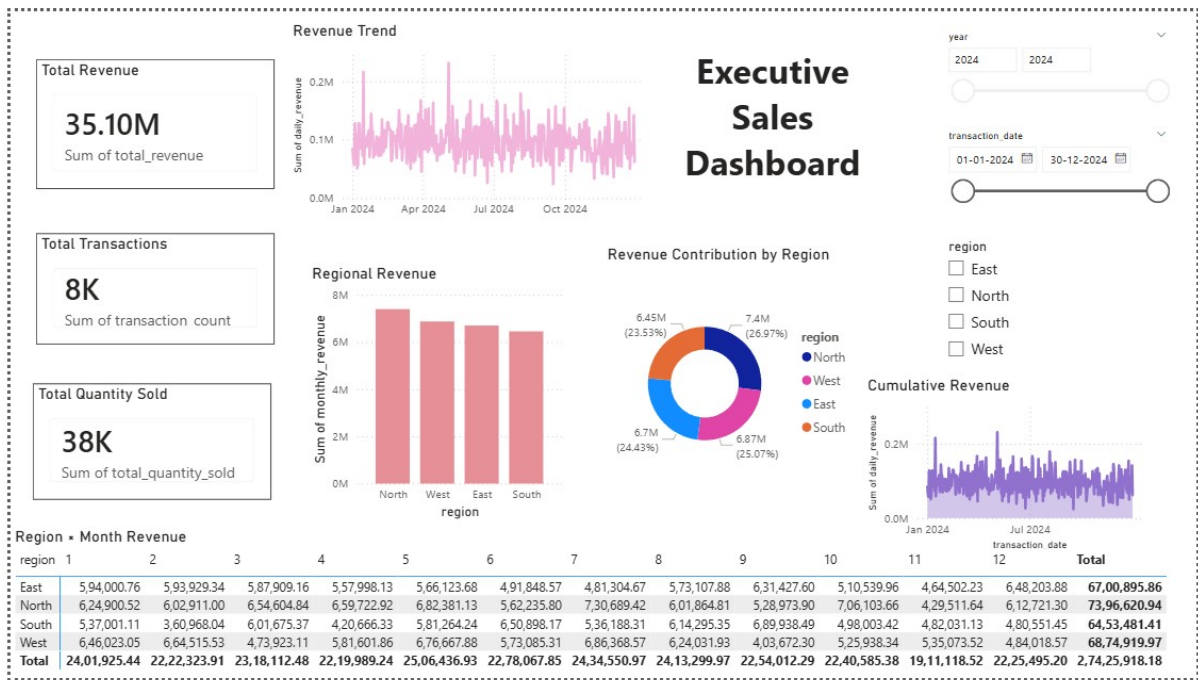


Power BI Dashboards

Gold tables are consumed in **Power BI** via Databricks SQL Warehouse to build business dashboards.

Dashboards Developed

- Executive Sales Dashboard
- Regional Performance Dashboard
- Product Performance Dashboard

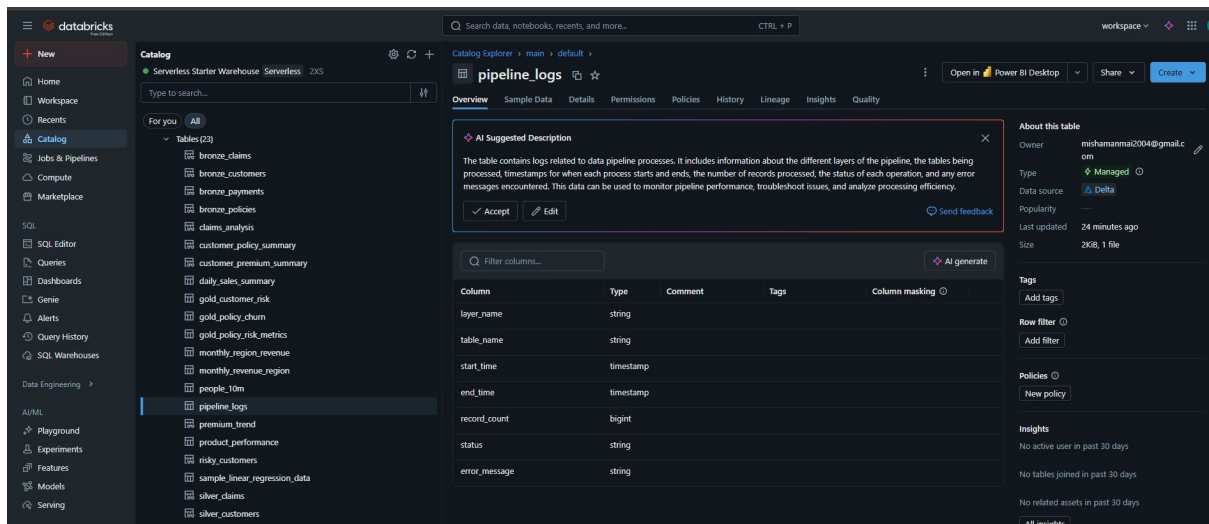


Pipeline Logging & Monitoring

Pipeline execution is logged to a dedicated Delta table to ensure observability and auditability.

Logged Information

- Pipeline start time
- Pipeline end time
- Record counts
- Execution status
- Error messages (if any)



7. Validation Summary

The following validations were successfully completed:

- Bronze ingestion preserves raw schema
- Silver layer enforces data quality and calibration
- Gold tables correctly aggregate business metrics
- Unity Catalog lineage verified for all Gold tables
- Power BI dashboards refresh successfully
- Pipeline logs confirm successful execution